

A HUMAN-LIKE AGENT FOR SCREEN-AWARE TASK AUTOMATION

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Topic	A Human-Like Agent for Screen-Aware Task Automation			
Project ID	25-26J-477			
Specialization	SE			
Student ID	IT22125248	IT22099518	IT22167378	IT22083296
PanelName	SE-P3			
Detailed Comments	She struggled to demo the live version - did not feel they have tested the demo workflow well. Even when switching to pre-recorded video, the overall message they were trying to convey was not clear. Did not show overall workflow going into multiple screens as highlighted on PP1.	She was not able to smoothly explain the work she did in brief. Took several questions to understand the work. Should demonstrate with more complex multiple screens in the overall workflow (as highlighted in PP1).	Clearer explanations of the features she did compared to the first two speakers. Did not show overall workflow going into multiple screens as highlighted on PP1.	Clearer explanations of the features she did compared to the first two speakers. Did not show overall workflow going into multiple screens as highlighted on PP1.

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Actions we took :

- Based on the feedback received during PP2, several improvements were implemented to enhance both the technical demonstration and presentation quality of the project. Enhanced the multi-screen workflow visualization and demonstrated the complete perceive-plan-act-evaluate cycle during testing sessions. Additionally, 30 functional test cases were executed to evaluate the reliability, accuracy, and adaptability of the proposed screen-aware automation system under different UI interaction scenarios.

Test Case Document Link :

https://drive.google.com/file/d/1cZv7Ba9JGI7uUx8LWGHIFIT1xem7gQW-/view?usp=drive_link

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Problem

- 60–80% of legacy systems lack automation access
- Traditional tools (Selenium, Appium) require internal system access
- Manual testing increases time, cost, and inconsistency

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Design Excellence over Existing Solutions

- Works in black-box environments (no internal access needed)
- Combines LLM reasoning + visual context
- Produces structured, executable outputs
- Supports adaptive re-planning
- Enables natural language-based automation



Agentic AI Logic



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- Acts as the central coordinator of the system. It manages task flow, integrates outputs from all modules, controls execution, and maintains the closed-loop process (perceive → plan → act → evaluate).

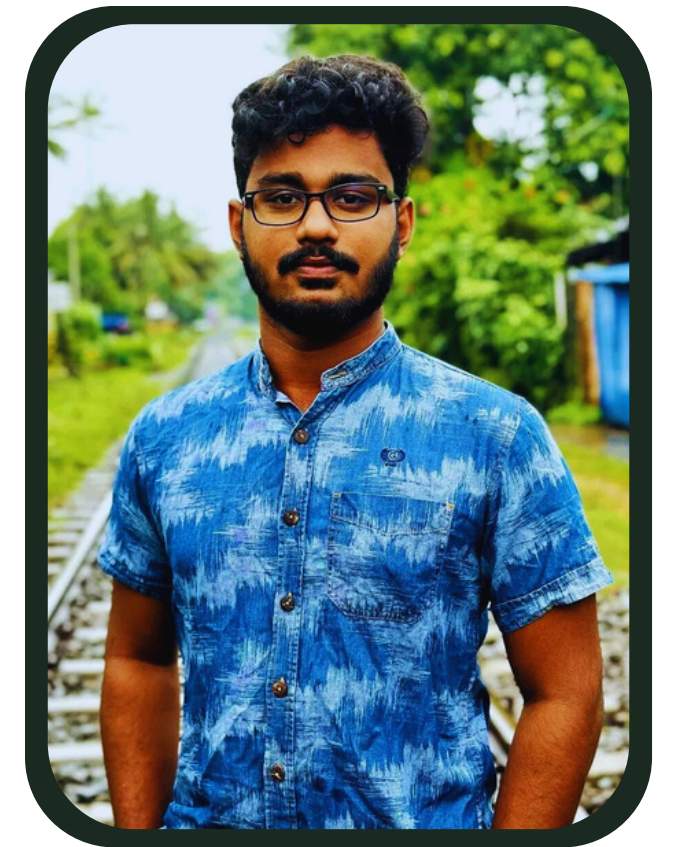
Visual Perception



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- The vision component captures and analyzes the screen using computer vision and OCR to detect UI elements, read text, and provide visual context for the agent.

HID Actuation



- The HID component translates action commands into low-level keyboard and mouse operations through ESP32-based USB HID emulation.

LLM & Prompt Engineering



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- The LLM component converts natural-language instructions and visual data into structured, validated action steps that the system can execute safely.

Test Cases Results:

Task Name	Type	Steps	Time	Result
Notepad Typing	Simple	1	12s	Success
Screen Analysis	Medium	1	5m 36s	Success
Login System	Complex	2	13m 39s	Failed
Button Navigation	Complex	2	17m 58s	Failed
Minimize & Open VS Code	Medium	2	12m 52s	Failed
Minimize & Open Chrome	Medium	2	9m 50s	Failed
GitHub Navigation	Complex	2	15m 03s	Failed
CourseWeb Login	Medium	2	9m 14s	Failed
Credential Login	Complex	2	18m 11s	Failed
YouTube Navigation	Medium	2	12m 19s	Failed
Create Repository	Complex	2	16m 11s	Failed
Open Repositories Tab	Simple	1	2m 47s	Success
Open GitHub Tab	Simple	1	7m 44s	Success
Open YouTube	Medium	2	42s	Success
Find Privacy Settings	Medium	2	1m 10s	Failed
Open Repo & Pull Requests	Medium	2	3m 28s	Failed
Gmail Search Invoice	Medium	2	3m 41s	Failed
Portal Login	Complex	3	1m 28s	Success
Download Installer	Medium	2	2m 16s	Failed
GitHub Issue Filter	Complex	3	5m 55s	Failed
Spreadsheet Copy Task	Complex	3	7m 04s	Failed
Chat App Upload	Complex	3	4m 33s	Failed
Rename File	Simple	3	4m 49s	Success
ClickUp Inbox Check	Complex	2	12m 51s	Failed
Gmail Open Email	Medium	2	3m 38s	Success
Enable Safe Browsing	Medium	2	4m 01s	Failed
GitHub Issue Open	Complex	3	4m 10s	Success
Calendar Check	Simple	2	3m 22s	Success
Move PDF File	Medium	2	5m 14s	Failed

Category	No. of Tasks	Completed	Failed	Success Rate
Simple	6	6	0	100%
Medium	12	8	4	66.7%
Complex	13	5	8	38.5%
Total	31	19	12	61.3%

Limitations:

- Performance decreases in complex multi-screen workflows.
- System accuracy depends on the quality of visual element detection.
- Limited contextual memory across long interaction sequences.
- Sensitive to unclear or poorly phrased user instructions.

Future Improvements:

- Introduce advanced state-memory and workflow tracking mechanisms.
- Improve multi-screen and long-horizon task planning.
- Support cross-platform environments including mobile and web systems.

**THANK
YOU.**